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Graded Assignment • 20 min

Due Jul 10, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Each student has a password, which is six characters long, and each character is either a digit or a lowercase letter. How many possible passwords are there?

1 / 1 point

- 26^6
- 26×6
- 36^6
- 36×6

Nice work

It is correct. There are six positions and we can fill in it by 36 items. So, it is 36*36*36*36*36*36=36^6.

2. Assume we have a list of two sets of items. In this case, the sum rule only applies when:

1 / 1 point

- The lists are identical.
- The lists partially overlap or share some common elements.
- The lists are pairwise disjoint.
- The lists are equal.

Nice work

Correct. Because if it is not pairwise disjoint then we should deduct the common items.

3. The Pigeonhole Principle states that if you have n+1 objects and n boxes in which to place those objects then:

1 / 1 point

- There is at least one box containing > 1 objects.
- Each box will contain only one object.
- Each box could contain any given number of objects.
- Each box will contain $\leqslant 1$ object.

Nice work

Correct. Watch The Pigeonhole Principle video.

4. Assume we have a family that consists of {father, mother, son, daughter} and we wish to take photos of them. In this case, we only have room in the photograph for pairs of two individuals, one in position 'A' and one in position 'B'.

How many permutations of photograph can we take to form a unique arrangement (e.g., A,B is one unique example and B,A is another unique example).

1 / 1 point

- 12
- 8
- 4
- 16

Nice work

Correct. It is P(4,2) which equals to 4!/2!=12.

5. In a class of 10 students, the teacher wants to select a group of 3 students to represent the class in a science competition. How many different groups of 3 students can be selected?

1 / 1 point

- $\binom{10}{7} = 720$
- $\frac{10!}{3!(10-3)!} = 120$
- $10^3 = 1000$
- $10/3$

Nice work

Great job! This is the correct formula for combinations where order does not matter.